CHAPTER 1 · LESSON 1 OF 5

What Is Data?

BY THE END

You can explain what data is, and tell data, information, and insight apart.

Overview

You check your phone and see a mobile money transfer of $25, a weather app showing 32°C, and a message from school about your attendance.



$25

a mobile money transfer of $25



32°C

a weather app showing 32°C



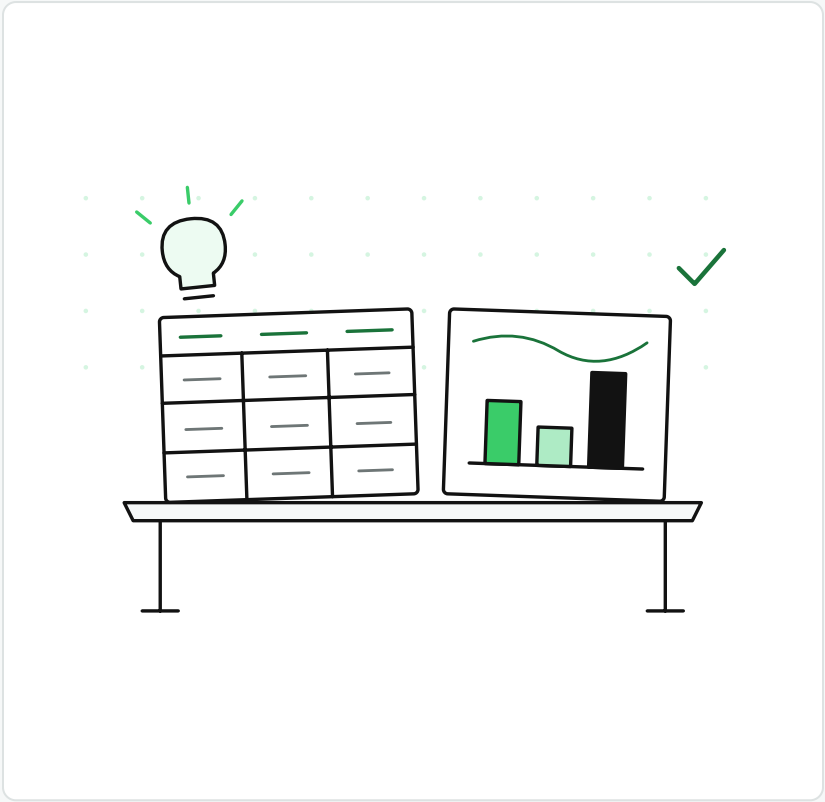
attendance

a message from school about your attendance

Each of those is a piece of data, a small fact about something that happened. Before you analyze data, build dashboards, or choose a career in data, you need a clear answer to a simple question: what is data?

Data: raw facts about the world

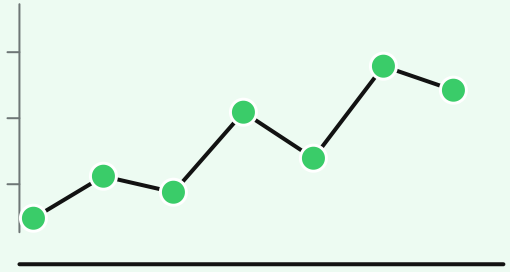
Data is a collection of facts, measurements, or observations about the world. On its own, data usually does not tell a full story. It is the raw material we use to understand what is happening.



Examples of data:

A mobile money transfer amount	`$25`
A market price for rice	`$2.50 per kg`
A student's attendance record	`present`
A product price	`$49.99`
A temperature reading	`32°C`

Each example is a single fact. Collect enough of them and patterns start to appear.



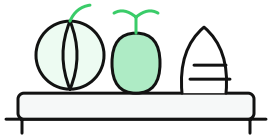
Data vs information vs insight

These three terms are related, but they are not the same.

TERM	MEANING	EXAMPLE
Data	Raw facts	Sales were $1,200, $980, and $1,450 on three days
Information	Data organized to answer a basic question	Sales increased from Monday to Wednesday
Insight	A useful conclusion that can guide action	Promotions on Wednesday may be driving higher sales

Data → Information → Insight

Think of it like cooking:



Data

ingredients

→



Information

a prepared meal you can understand

→



Insight

knowing what to cook next time based on what people liked

Where you see data every day

Data is not only found in spreadsheets or databases. You create and use data constantly:

Mobile money transactions and bank transfers

School grades and attendance records

Market prices at a local shop

Likes and comments on social media

Steps tracked by a fitness app

Weather forecasts

Businesses, schools, hospitals, and governments all rely on data to make decisions.